Survival Analysis UB Learns Open Book Quiz

Question 1: When to Use Survival Analysis
Which of the following study designs would **most appropriately** require survival analysis techniques?
A) A cross-sectional survey of current smoking prevalence
B) A case-control study of past infection and cancer
C) A longitudinal study measuring time to hospitalization
D) A randomized trial comparing cholesterol levels at 6 months

Answer: C) A longitudinal study measuring time to hospitalization
Explanation: Survival analysis is used when the outcome is **time until an event**, and not just event status or levels of a continuous variable.

Question 2: Interpreting the Hazard Function
A clinical trial compares a new chemotherapy regimen with standard treatment in patients with advanced lung cancer. The hazard function for the new treatment group remains flat over time, while the control group's hazard increases steadily.
Which of the following best describes the implication of this hazard pattern for the treatment group?
A) Patients are at increasing risk of failure over time
B) Patients are at constant risk of failure over time
C) The treatment leads to delayed onset of risk
D) There is no failure observed in the treatment group

Answer: B) Patients are at constant risk of failure over time
Explanation: A flat hazard function implies a constant risk per unit time, which is characteristic of the exponential distribution and often assumed in simple survival models.

Question 3: Censoring Mechanism
In a survival study of kidney transplant patients, some participants are lost to follow-up, and some are still alive at the end of the study period.
What type of censoring is this most consistent with?
A) Left censoring
B) Interval censoring
C) Informative censoring
D) Right censoring

Answer: D) Right censoring
Explanation: Right censoring occurs when the event of interest has not occurred by the end of the observation period or when a participant is lost to follow-up.

Question 4: Informative vs. Non-informative Censoring
In a survival analysis of patients with HIV, those who become too ill to return for follow-up are censored.
Which type of censoring does this represent?
A) Right censoring
B) Non-informative censoring

C) Informative censoring
D) Left censoring

Answer: C) Informative censoring
Explanation: If the reason for censoring is related to the likelihood of the event (e.g., being too ill), it violates the non-informative assumption.

Question 5: Left Censoring
Which of the following scenarios illustrates **left censoring**?
A) A patient is lost to follow-up before experiencing the event.
B) A patient enrolls in a study and has the event after 12 months.
C) A patient had the event before entering the study, but the exact time is unknown.
D) A patient did not have the event before the study ended.

Answer: C) A patient had the event before entering the study, but the exact time is unknown.
Explanation: Left censoring occurs when the event has already happened before observation begins, but its timing is not known.

Question 6: Interpretation of the Survivor Function
A survival function S(t)=0.70 at t=12 months in a cancer study indicates:
A) The event rate is 70% at 12 months
B) 70% of patients had the event by 12 months
C) The hazard at 12 months is 0.70
D) 70% of patients are expected to survive beyond 12 months

Answer: D) 70% of patients are expected to survive beyond 12 months
Explanation: S(t) represents the probability of surviving **beyond** time t.

Question 7: Survivor Function at t=0
What is the value of the survival function S(t) at time t=0, assuming no events have occurred at baseline?
A) 0
B) 0.5
C) 1
D) Undefined

Answer: C) 1
Explanation: At t=0, all individuals are alive or event-free, so S(0)=1.

Question 8: Nature of the Hazard Function
Which of the following best defines the **hazard function**?
A) The number of events per unit time
B) The total risk accumulated by time t
C) The instantaneous event rate at time t given survival until t
D) The proportion of censored observations at time t

Answer: C) The instantaneous event rate at time ttt given survival until ttt
Explanation: The hazard function captures the conditional risk of failure in a tiny interval, given survival up to that time.

Question 9
Which of the following best describes the **main goal** of survival analysis in clinical research?
A) To estimate the correlation between exposure and binary outcome
B) To compare means across two treatment groups
C) To estimate and compare the time until occurrence of a specific event
D) To maximize sensitivity and specificity of a diagnostic test

Answer: C) To estimate and compare the time until occurrence of a specific event
Explanation: Survival analysis focuses on **time-to-event** data, accounting for censoring and the distribution of event times.

Question 10
In a cancer trial, a patient withdraws consent and stops all follow-up 10 months into a 3-year study, with no event observed up to that point. How should this patient's data be handled in the survival analysis?

A. Excluded from the analysis entirely, since their eventual outcome is unknown
B. Treated as an event at 10 months, since that's when they left the study
C. Censored at 10 months, contributing at-risk time up to that point
D. Censored at 3 years, the end of the study

Answer: C) Censored at 10 months, contributing at-risk time up to that point
Explanation: Right-censoring means follow-up ended before the event of interest occurred. The patient contributes at-risk time up to their last known point (10 months) but is not counted as an event, and is not simply dropped from the dataset.